Your simulations: keys

WyomingWill's demos in bpp: the rec room and its games, the clocks, Euler's Cabinet of Curiosities, artificial physics and the walkers · Part 2 of 2 (Part 1 is bpp itself) · October 9, 2026

Click the 3D view first so it has keyboard focus. Each demo lists its own keys in the Shortcuts pane. A key a demo uses wins over bpp's own meaning for it (Part 1 lists those). Games keep the same keys in the rec room as when they run by themselves.

Rec room

`Tab`	Walk to the next spot: pinball machine → pool table → snooker table → bumper pool table → clock → marble run → walker → whole-room view → back to pinball
`Shift+Tab`	Walk the other way
`Mouse`	Turn and zoom the view (anywhere)
`S D R P`	At a game: only what that game uses them for (the tables' S and D are spin, the clock's S is its sound); nothing at a game that doesn't use them. In the room view: bpp's usual meaning (stop, sleeping off, reload), except P, which is held back (POV-Ray export would slow the room to about 2 frames a second).

Walking away from a game lets go of any keys you're holding (a raised flipper drops, a drawn plunger launches); the game carries on. A game may show a **PAUSED** sign when the room is too busy; it picks up again by itself when there's room. The clock is never paused.

Pinball machine

Playing

`Return`	Hold to pull the plunger, release to launch (with no game running, launching starts one)
`Left / Right Shift`	Flippers (Z and / also work)
`Space`	Shake the machine (a push up the table). Too often: DANGER, then TILT, and the ball is lost
`1`	New game (when no game is running)
`P`	The computer plays, learning as it goes (P again to stop)
`V`	The view: the player's, or the whole machine
`R or Ctrl+R`	Reload the table (after editing pinball-machine-a-rules.lua)

Layout editor

The editor is open when the table loads (or is reloaded) and closes when you press E or launch the first ball. Select a part, then move it with the arrows.

`L`	The rollover lanes (as a group)
`1 2 3 …`	That pop bumper (1 left, 2 right, 3 bottom, 4 and up: added ones)
`B`	Add a pop bumper (up to 9); it appears selected
`Delete`	Remove the selected added bumper
`A`	The top arch: Up/Down raise or lower it (the lanes come with it), Left/Right shift its flat top
`O`	The orbit entrance: inner wall, post and targets

X	The orbit exit: over the inlane returns balls, over the outlane drains them
F	Flippers, inlane guides and slingshots (together)
M	The ramp: arrows move its entrance; , and . turn it 2.5°
Arrows	Move the selection: tap 0.5 cm, hold to slide (speeds up)
0	Put the selection back to the default layout
R	Reset the whole layout to the default (your saved layout is kept until you press E)
E	Finish editing and save the layout

Pool table

8-ball against the computer, or clearing the table alone.

Left / Right	Aim: tap for a quarter degree, hold to swing
, .	Fine aim (1/50°)
Up / Down	Force (1% a tap)
W / S	Hit higher (follow) / lower (draw)
A / D	Side: left / right
C	Centre hit
E / Q	Raise / lower the cue (massé, jump shots)
Space or Return	Shoot
G	Aiming guide on/off
V B T	Camera: starting view, behind the cue, overhead
N or R	Re-rack
M	Switch game: 8-ball against the computer, or clear the table alone
P	Auto-play: the computer plays both sides, rack after rack

Ball in hand

Arrows	Move the cue ball (Up = away from you)
Space or Enter	Put it down

Snooker table

Aiming, force, spin, cue height, shooting and cameras are the same keys as the pool table. The differences:

G	Aiming guide; the ghost ball turns red when the ball you'd hit first isn't on
N or R	New frame
M	Switch game: a match against the computer, or break building alone
P	Auto-play: the computer plays both sides, frame after frame
Arrows, Space	Ball in hand: move the cue ball around the D, then put it down

Bumper pool table

Aiming, force, spin (W A S D C), E/Q, Space/Return to shoot and the cameras V B T are the same as the pool table. The differences:

X / Z	Next / previous of your balls (the ring shows which)
Tab	Same as X, when the table runs by itself (in the rec room Tab walks on)
G	Aiming guide: to the first cushion, bumper or ball
N or R	New game
O	White: the computer, or a second player
L	The computer's level: 1 easy, 2 medium, 3 hard
P	The computer plays red too

Clock

T	Start gravity training (tunes gravity toward 60 s for one turn of the second hand)
G	Lock gravity at its current value and stop training
S	Tick-tock sound on/off
Slider	**clock_gravity** in the Params pane (moves the clock alone)

Marble run and walker

(no keys)	Their settings are sliders in the Params pane, named after them (**marble_run_…**, **walker_…**, e.g. walker_Speed). At the walker the view follows it. Either is put back as it started if a part leaves its table or floor.

The clocks (run by themselves)

N2_Clock_handcheck, N7_Clock_handcheck and E3_Clock_handcheck in the Clocks folder. The rec room's clock is one of these, with the same keys.

T	Start tuning toward 60 s a minute (N2 and N7: the gravity; E3: the foliot weight)
G	Lock the current value and stop tuning
S	Tick-tock sound on/off

Euler's Cabinet of Curiosities

Rattleback

R	Spin all three again
T	Stop them, then tap all three on the end
Z	Slow motion (1/4) ↔ real time

Mouse-over: rest the pointer on one for half a second to see which way it is spinning and what it will do.

Dzhanibekov effect

R	Start again
Z	Slow motion (1/4) ↔ real time

Mouse-over: rest the pointer on one for half a second to see which axis it is spun about and whether that is the smallest, middle or largest moment.

Gomboc Drop C and Gomboc Variety

R	Drop them at random orientations
U	Stand them on their heads
K	Kick them up with a spin
] / [	One more / one fewer

Mouse-over (Gomboc Variety): rest the pointer on one for half a second to see its shape, size, mass, tilt and state.

Bille Drop

R	Drop them at random orientations
U	Set them down on face A, B or C
K	Kick them up with a spin
Z	Slow motion: 50× ↔ 10× slower
] / [	One more / one fewer

Artificial physics

Formations: Lennard-Jones and Newtonian (the same keys)

S	Start/stop the simulation
A	World axis on/off
R	Set up again: re-scatter the agents and the goal
2 / 3	Switch to 2D / 3D
] / [	Add / remove a particle
D	Disable a random active particle
F	Square formation on/off
G	Goal attraction on/off
C	Clear all obstacles
Left/Right or J/L	Move the cursor along x
Up/Down or I/K	Move the cursor along z
Y / H	Move the cursor along y (3D only)
O	Drop an obstacle at the cursor
. / ,	Faster / slower playback (delayMs)
= / -	More / fewer ticks per frame
V	Neighbour-connection lines on/off

Uniform Coverage

] / [	Five more / five fewer spheres
. / ,	Faster / slower playback (delayMs)
= / -	More / fewer ticks per frame
S	Start/stop the simulation

Walkers and the Hover example

F1 / F2	Walking_Linkages.lua: previous / next mechanism
(no keys)	The single walkers in the Walkers folder (Chebyshev, Hoecken, Jansen, Klann, Spears) and the Hover example: sliders only, in the Params pane. In the Hover example, rest the pointer on an object to see what it is and its live position and speed.

Walker room

`Tab`	Follow the next walker (the view moves along with it; the mouse still turns and zooms it)
`Shift+Tab`	Follow the one before
`O`	Look over the whole room
`T`	Clear the trails
`(slider)`	TerrainHeight, in the Params pane: how high the bumps are, 0 to 8 cm. BumpLength: how long they are, 11 to 44 cm (every walker's stride is 15 cm); short bumps are obstacles a foot has to get over. The room is built again, every walker back at its start, when you let go.
`(pointer)`	Rest it on a walker, or on a trail, to see whose it is, how far it has gone and how often it has turned or been stood up.